

TED (10)- 5028

Reg No.....

REVISION -2010

Signature.....

MODEL QUESTION PAPER

SIXTH SEMESTER DIPLOMA EXAMINATION IN MECHANICAL ENGINEERING

5028 MAINTENANCE ENGINEERING

(Maximum mark :100)

(Time:3 hours)

PART - A

I) Answer the following question in one or two sentence. Each question carries 2 marks

1. Define maintenance
2. What is M.T.T.R?
3. Define frequency of vibration
4. What is condition monitoring?
5. List two method of corrosion monitoring

II) Answer any *five* following. Each question carries 6 marks

PART - B

1. Compare planned maintenance and routine maintenance
2. Explain a lubrication chart
3. What are the factors of availability
4. Explain proximity analysis
5. Explain Radiographic testing
6. Explain pistol thermometer method of condition monitoring
7. What are the factors leading to the development of failure.

PART - C

(Answer *one* full question from each unit. Each question carries 15marks)

Module-I

- II) (a). Explain the various application of Computers in maintenance. (8)
- (b). List the advantages of preventive maintenance (7)

OR

- IV). (a). Explain the flow chart for executing preventive maintenance (8)
- (b). What are the principles of lubrication? (7)

Module-II

- V) (a). Explain reliability, maintainability and availability (8)
(b). Failure rate = 0.001 unit/hr, what is the reliability for 800 hrs. (7)

OR

- VI) (a). Discuss a typical Bath Tub curve (8)
(b). Two detection systems are used and both of them are able to work and each system reliability (7) reliability is 0.9. Find the

Module-III

- VII) (a). Explain the effects of misalignment on the shafts. (8)
(b). Explain thermography (7)

OR

- VIII) (a). Explain vibration limits and severity criteria (8)
(b). What is vibration signature (7)

Module-IV

- IX) (a). Explain failure Tree Analysis
(b). Explain Radiographic testing

OR

- X) (a). What is N.D.T? Explain the principle of ultrasonic testing (8)
(b). Explain the importance of corrosion monitoring (7)